ME 1600 – Python Project

Section 1: due on Dec. 7th 9:55 AM via Canvas
Section 4: due on Dec. 7th 2:15 PM via Canvas

Background Theory

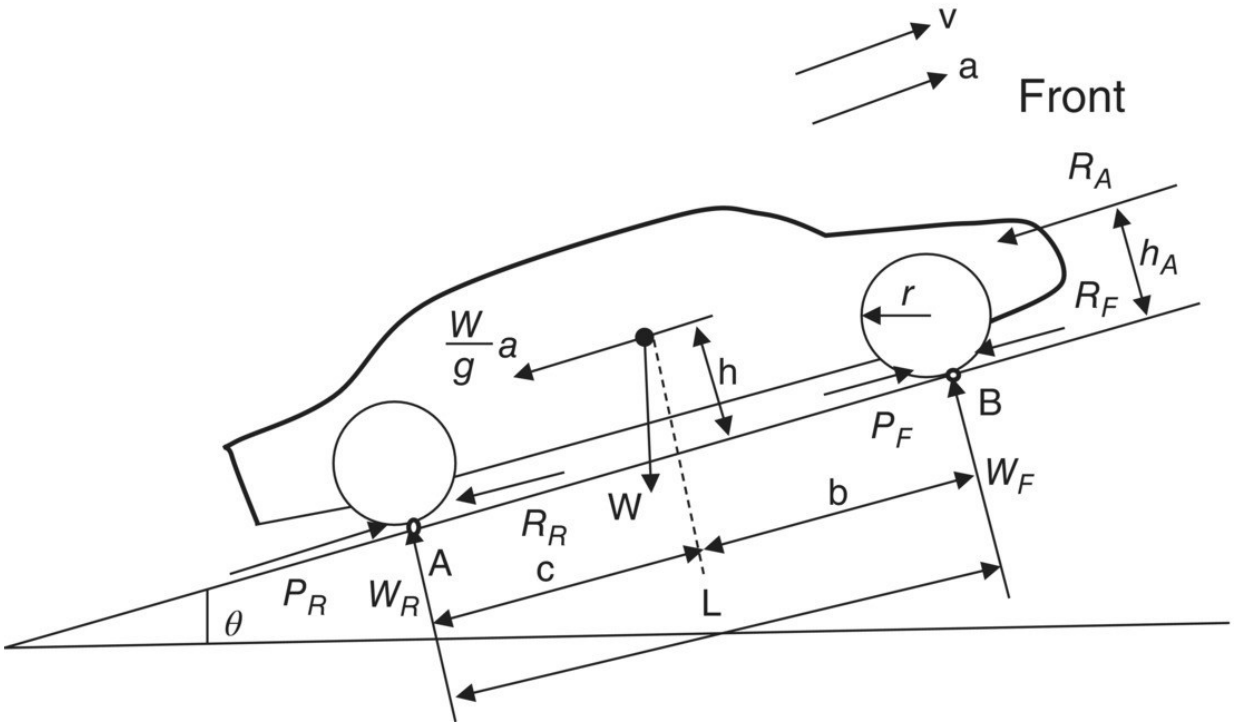


Figure 1 – Free Body Diagram.

Acceleration and Axle Loads. In Fig. 1, *A* and *B* denote the points of contact between the wheels and the road surface. θ is the *angle of the slope in radians*, and *r* is the *radius* of the tire. The *height of the center of gravity* and the *air resistance height* are denoted as *h* and h_A. For passenger cars, these two heights are assumed to be the same. The vehicle *wheelbase* is *L,* and the *longitudinal position of the gravity center* is determined by *b* and *c*, which are the distances from the gravity center to the front axle and rear axle, respectively. W_F and W_R, called *axle loads*, are the contact forces between the front wheels and the road surface and between the rear wheels and the road surface. Particularly, *static axle weights on level ground* are obtained from the moment balance equations by making the slope angle θ, the air drag R_A, and the acceleration *a* equal to zero.

Under several assumptions (D'Alembert's principle), we can transform an accelerating rigid body into an equivalent static system by adding the so-called "inertial force". The inertial force must act through the center of mass. The system can then be analyzed exactly as a static system subjected to this "inertial force" and the external forces. In Fig. 1, Wa/g is the *inertia force*, where *W* is the *vehicle weight*, *a* is the *acceleration*, and *g* is the *gravitational acceleration*.

The forces in the free body diagram form a system of equilibrium, and three scalar equations can be written based on the condition of equilibrium:

$$\sum M_A = 0 \tag{1}$$

$$\sum M_B = 0 \tag{2}$$

$$\sum F = 0 \tag{3}$$

The first two equations are based on the condition that the sum of moments made by all forces about point A and point B must be equal to zero. The moment of a force is a measure of its tendency to cause a body to rotate about a specific point or axis. The magnitude of the moment of a force acting about a point or axis is directly proportional to the distance of the force from the point or axis. It is defined as the product of the force and the moment arm. The moment arm or lever arm is the perpendicular distance between the line of action of the force and the center of moments.

The third equation above is that the sum of all forces, including the inertia force, must be equal to zero in the direction of vehicle motion. These equations can be arranged to express the axle loads W_F and W_R, which can be evaluated after calculating a, P and R (see below).

Road Load. *Road load* (R) is against the vehicle's motion, while *traction* (*P*), which originates from the engine, propels the vehicle. Note that in this discussion, with *P* we refer to the *total traction* that propels the vehicle, not the force applied to each driving wheel.

The *road load* (R) is the resultant of three separate forces: *rolling resistance* ($R_{rolling}$), *grade load* (G_L), and *air resistance* (R_A).

$$R = R_{rolling} + G_L + R_A \tag{4}$$

In Fig. 1, R_F and R_R are the *rolling resistances* from the front and rear wheels respectively, so that:

$$R_{rolling} = R_F + R_R = fW\cos\theta \tag{5}$$

where W is the *weight* of the vehicle, and the angle θ depends on the *road slope*. f is called *rolling resistance coefficient,* and depends on many factors, such as tire material, texture, tread, inflation, speed, etc.

The *grade load* (G_L) is the *component of the weight W* along the vehicle direction.

Finally, R_A is the *air drag* or *air resistance*. The air drag is a distributed load, but for simplicity it is assumed to be acting at height h_A. There are two causes for the generation of air resistance: friction between the air and the vehicle body surface and air turbulence formed around the vehicle body. The latter is the main cause of air drag for ground vehicles. Factors affecting the magnitude of the air drag include the shape and finish of the vehicle body, the vehicle's frontal projected area, the air density, and most importantly, the vehicle's speed:

$$R_A = \frac{1}{2}\rho C_D A(v)^2 \tag{6}$$

where v is the *vehicle relative speed*, C_D is the unitless air *drag coefficient* that mainly depends on vehicle body shape and body surface smoothness, ϱ is the *air density*, and A, the *vehicle frontal projected area*, is the vehicle body area that confronts the airflow in the direction perpendicular to vehicle motion.

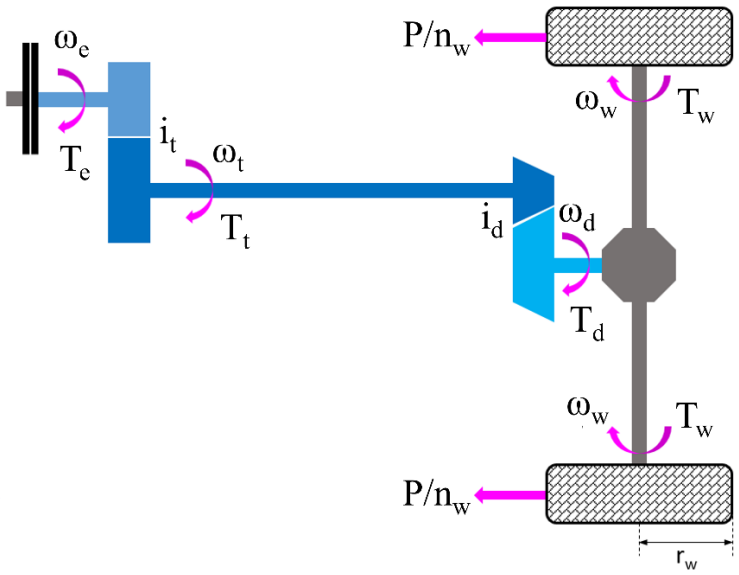


Figure 2 – Layout of MT (Manual Transmission) powertrain.

Traction and Powertrain. The *tractions* P_F and P_R (Fig. 1) are available from both front and rear wheels only for a *four-wheel drive* vehicle (4WD). In a front-wheel drive (FWD) vehicle, the traction is primarily at the front wheels, so that P_R is zero and P_F is equal to the total traction P. This means that the engine's power is transmitted to the front wheels, which are responsible for both driving and steering the vehicle. This setup is common in many passenger cars due to its cost-effectiveness and efficient use of space. Analogously, in a rear-wheel drive (RWD) vehicle, the traction is primarily at the rear wheels, so that P_F is zero and P_R is equal to the total traction P. In a rear-wheel drive vehicle, the engine's power is transmitted to the rear wheels, which are responsible for propelling the vehicle forward. This setup is often found in sports cars and trucks, as it can provide better balance and handling, especially under acceleration.

If a two-wheel drive vehicle (FWD or RWD) is driven on a straight line, the *torque at the differential* T_d (Fig. 2) is equally split between the two driving wheels so that the *torque at the wheel* T_w is:

$$T_W = \frac{T_d}{n_w} \tag{7}$$

with n_w the *number of driving wheels.*

Additionally for *each* driving wheel, with P the *traction* and r the *wheel radius:*

$$T_W = \frac{P}{n_w}\, r \tag{8}$$

So that, ultimately, combining equations 7 and 8:

$$P = \frac{T_d}{r} \tag{9}$$

Therefore, for evaluating P, it is necessary to analyze the entire power train system (Fig. 2) in which the engine output is transmitted through the transmission to the drive shaft, which then transmits the transmission output to the final drive through a pair of spiral bevel gears. In this discussion, the *engine*

torque and *engine angular velocity* are denoted as T_e and ω_e respectively, the *transmission torque* and *transmission angular velocity* are denoted as T_t and ω_t, and the *angular velocity of the driving wheel* is denoted as ω_W.

When a vehicle travels on the road, there is always a small amount of slippage between the tire and the road surface. However, under normal driving conditions, the slip rate is very small and can be neglected for the selection of gear ratios. If the slippage is not considered, then vehicle motion and driving wheel rotation are related as:

$$\omega_W = \frac{v}{r} \tag{10}$$

Assuming again that the vehicle proceeds in a straight line, at the drive gearset:

$$T_d = \eta_d i_d T_t \tag{11}$$

$$i_d = \frac{\omega_t}{\omega_W} \tag{12}$$

where i_d is the *final drive ratio* (defined as the division of the input angular velocity by the output angular velocity) and η_d is the *final drive efficiency*.

Similarly, the outputs of the engine and the transmission are related by the following equations:

$$T_t = \eta_t i_t T_e \tag{13}$$

$$i_t = \frac{\omega_e}{\omega_t} \tag{14}$$

where i_t is the *transmission ratio*, and η_t is the *transmission efficiency*. The transmission ratio is a stepped variable for manual transmissions. For a five-speed transmission, the ratio varies between the highest value in the first gear and the lowest value in the fifth gear (Fig. 3).

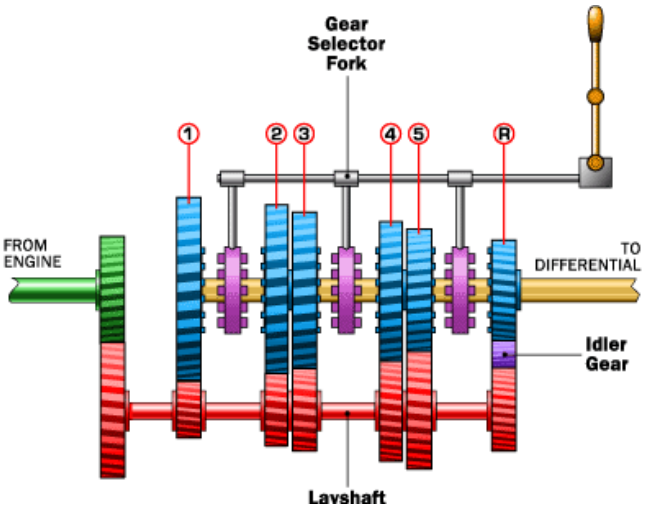


Figure 3 – 5-speed manual transmission geartrain

These considerations directly relate the traction P to T_e (via equations 9, 11 and 13) so that the task becomes to evaluate the engine torque.

Engine Torque and Power. The engine torque is often plotted as a curve against the engine speed in RPM for specific throttle openings (as a percentage of the wide-open throttle, WOT, or, sometimes,

as a degree of throttle angle) as shown in Figures 4 and 5. These curves are specifically related to a particular engine, and they are found experimentally using a *dynamometer,* or "dyno" for short, which is able to simultaneously measure engine torque and rotational speed.

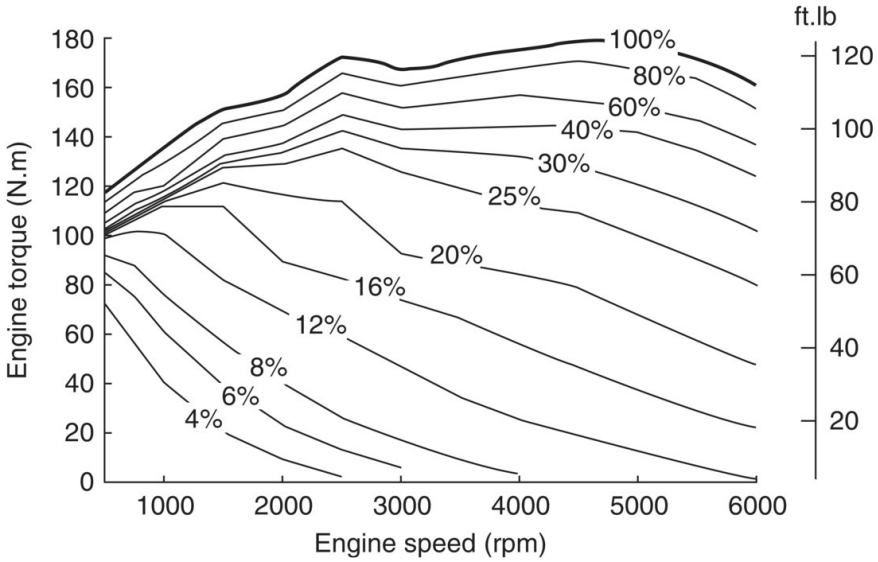


Figure 4 - Engine torque curves for various throttle openings.

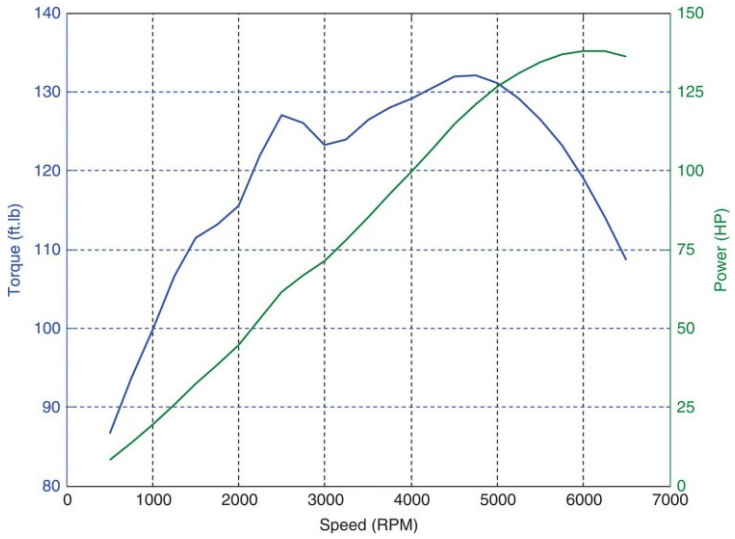


Figure 5 - Engine torque and power at wide open throttle.

Since the evaluation of ω_W is possible using equation 10, given speed and wheel radius, the calculation of ω_e is easily performed (via equations 12 and 14) and the experimental relationship finally leads to the estimation of T_e, and, therefore, P. This will lead to the calculation of the acceleration and, ultimately, the axle loads via equations 1, 2 and 3.

Additionally, the *engine power* then can be determined as:

$$Power = T_e \omega_e \tag{15}$$

Engine Fuel Map. Note that the fuel efficiency, a top performance specification in today's automotive industry, is also directly connected to ω_e and T_e. The fuel consumption data of internal

combustion engines are experimentally obtained by intensive engine mapping and are usually provided as a *fuel map*, which indicates in contour plots the specific fuel consumption of the engine at a given operation status, as shown in Figure 6. The specific fuel consumption is the amount of fuel that the engine needs to burn in order to do one horsepower-hour of work, either in liters, grams, or pounds of weight.

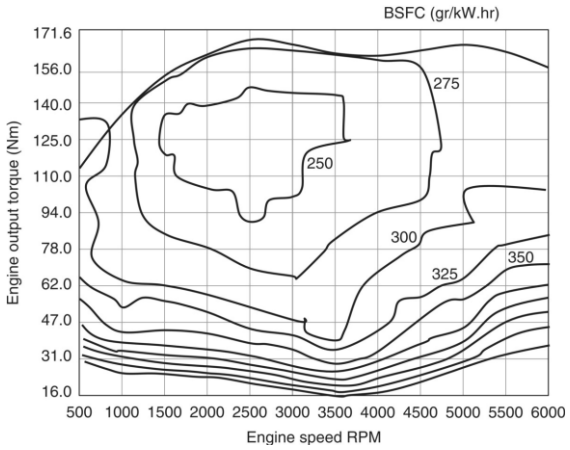


Figure 6 - Engine specific fuel consumption map.

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Yi Zhang, Chris Mi, "Automotive Power Transmission Systems," Wiley, 2021

Yong Chen, "Automotive Transmissions: Design, Theory and Applications," Springer, 2021

Robert Fischer, Ferit Küçükay, Gunter Jürgens, Rolf Najork, Burkhard Pollak, "The Automotive Transmission Book", Springer, 2015

Harald Naunheimer, Bernd Bertsche, Joachim Ryborz, Wolfgang Novak, "Automotive Transmissions: Fundamentals, Selection, Design and Application," Springer, 2011

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The program requirements

Write a Python modular program to analyze vehicle performance at wide-open throttle, proceeding on a straight trajectory. The user should be able to run this program repeatedly until actively exited. All the functions (mandatory) must be collected in a module (mandatory).

The program must:

- prompt the user about the type of vehicle analyzed (on-screen menu)
- import six gear ratio values from a given Excel file (see the provided example file). The program must also check that each gear ratio value entered is positive and prompt the user again if negative, saving the new value in the original Excel file.
- import all the other values from a given Excel file (see the provided example file): drive ratio (also called final drive) (i_d), transmission efficiency as % (η_t), final drive efficiency as % (η_d), radius of the wheel in m (r), speed in km/h (v), terrain slope as % (θ), weight of the vehicle in N (W), center of gravity height (h) in m, considered equal to the air resistance moment arm (h_A), wheelbase in m (L), distance identifying the center of gravity position from the front wheel in m (b), roll resistance coefficient (f), the air density (ϱ) in kg/m^3, the drag coefficient (CD) and the frontal area (A) in m^2. The program must also check that the values entered are positive and prompt the user again if negative, saving the new values in the original Excel file.
- import the dynamometer's experimental data from a given Excel file (see the provided example file): ω_e in RPM, T_e in N·m and ω_{MAX} in RPM.
- calculate for all the gear ratios provided: the engine angular velocity in RPM (ω_e), the engine torque (T_e using the best fit of the data file, order 4) in N·m, the power in horsepower (HP), the traction (P) in N, the road load (R) in N, the front axle weight static on level ground (W_{Fstat}) in N, the rear axle weight static on level ground (W_{Rstat}) in N, the acceleration in (a) in m/s^2, the dynamic front axle weight (W_F) in N, and the dynamic rear axle weight (W_R) in N.
- save in a new Excel file all the results and the data of the analysis. It creates the file if it is not there. The results should be saved in multiple tabs if the user runs multiple analyses. If the engine velocity exceeds ω_{MAX} RPM, all the cells in that line should contain just "OOR" for "out of range."
- plot different calculated variables in a (minimum) three-subplot figure as a function of the car velocity or other parameters, according to your curiosity/interest. Some plots must include multiple curves for different gear ratios (mandatory). The plot must exclude results for engine velocity exceeding ω_{MAX} RPM. The plots use the parameters passed for the simple analysis and just use some specific variables as arrays. Feel free to include more figures in addition to the three required.

Data for testing

2001 AP1 Honda S2000

CAR DATA

GEAR RATIOS	value	test
1st gear	3.133	3.133
2nd gear	2.045	2.045
3rd gear	1.481	1.481
4th gear	1.161	1.161
5th gear	0.971	0.971
6th gear	0.811	0.811

OTHER PARAMETERS	symbol	value	test
speed in km/h	v	100.000	100.000
terrain slope as %	slope %	2.000	2.000
wheelbase in m	L	2.400	2.400
radius of the wheel in m	r	0.406	0.406
roll resistance coefficient	f	0.018	0.018
air resistance moment arm, center of gravity height in m	hA,h	0.495	0.495
final drive efficiency as %	ηd	95.000	95.000
transmission efficiency as %	ηt	96.000	96.000
weight of the vehicle in N	W	12446.600	12446.600
air density in kg/m^3	ρ	1.230	1.230
drive ratio (final drive)	id	4.100	4.100
distance identifying the center of gravity position from the front wheel in m	b	1.540	1.540
drag coefficient	CD	0.330	0.330
frontal area in m^2	A	1.870	1.870

DYNAMOMETER DATA

ωe (RPM)	Te (N m)
1800.00	176.00
2200.00	185.20
2600.00	192.40
3000.00	199.30
3400.00	206.70
3800.00	214.90
4200.00	219.60
4600.00	222.20
5000.00	224.00
5400.00	228.30
5800.00	232.00
6200.00	234.30
6600.00	235.20
7000.00	234.00
7400.00	230.10

MAX ANGULAR VELOCITY

ωmax (RPM)	9000

2024 Lotus Emira

CAR DATA

GEAR RATIOS	value	test
1st gear	3.540	3.540
2nd gear	1.910	1.910
3rd gear	1.410	1.410
4th gear	1.030	1.030
5th gear	0.970	0.970
6th gear	0.860	0.860

OTHER PARAMETERS	symbol	value	test
speed in km/h	v	100.000	100.000
terrain slope as %	slope %	2.000	2.000
wheelbase in m	L	2.570	2.570
radius of the wheel in m	r	0.508	0.508
roll resistance coefficient	f	0.018	0.018
air resistance moment arm, center of gravity height in m	hA,h	0.490	0.490
final drive efficiency as %	ηd	95.000	95.000
transmission efficiency as %	ηt	91.000	91.000
weight of the vehicle in N	W	14288.400	14288.400
air density in kg/m^3	ρ	1.230	1.230
drive ratio (final drive)	id	3.800	3.800
distance identifying the center of gravity position from the front wheel in m	b	1.285	1.285
drag coefficient	CD	0.340	0.340
frontal area in m^2	A	2.020	2.020

DYNAMOMETER DATA

ωe (RPM)	Te (N m)
1000	180
1400	248
1800	304
2200	348
2600	383
3000	409
3400	427
3800	437
4200	442
4600	440
5000	432
5400	420
5800	404
6200	385
6600	364

MAX ANGULAR VELOCITY

ωmax (RPM)	7000

Possible Outputs

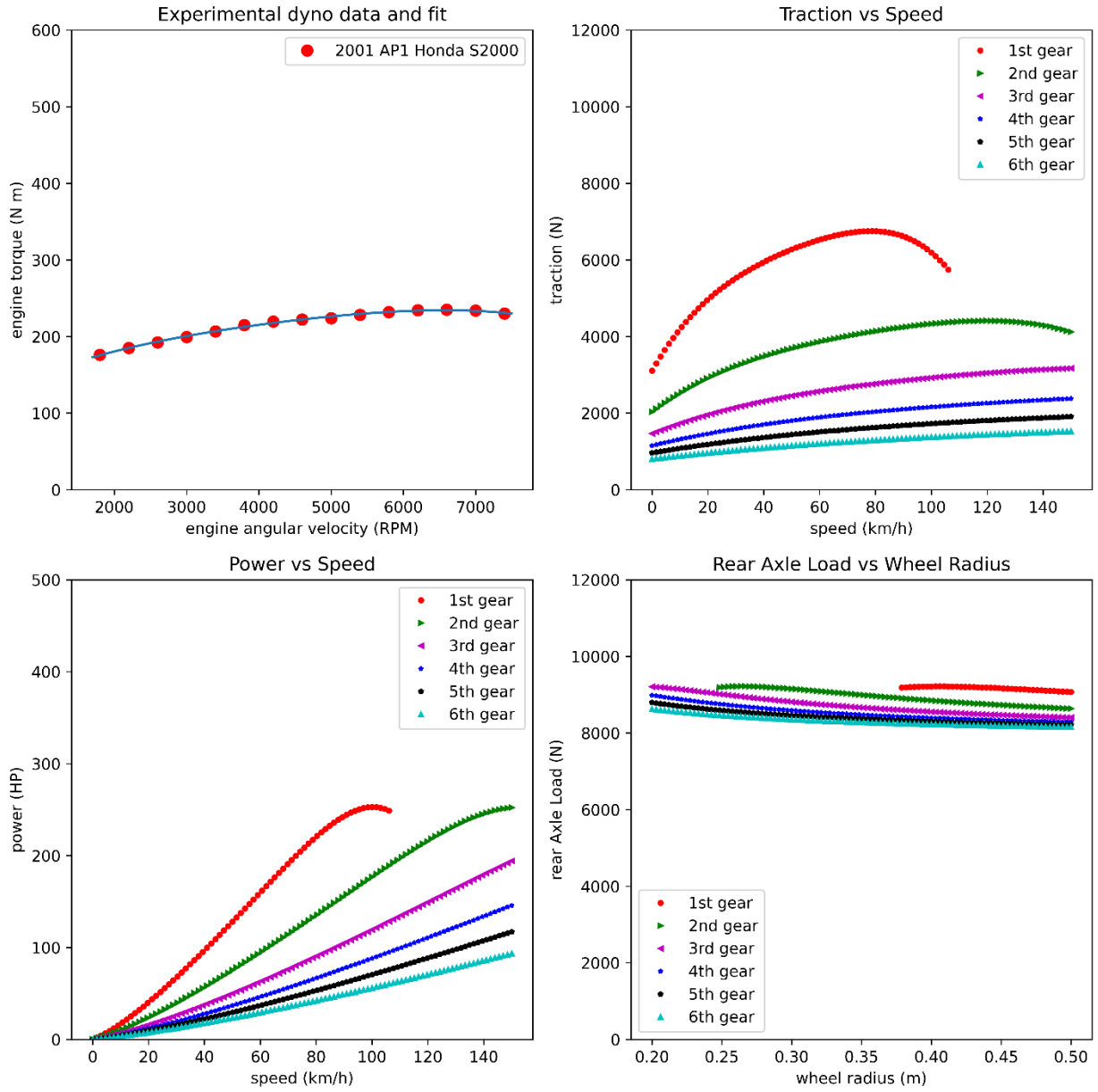
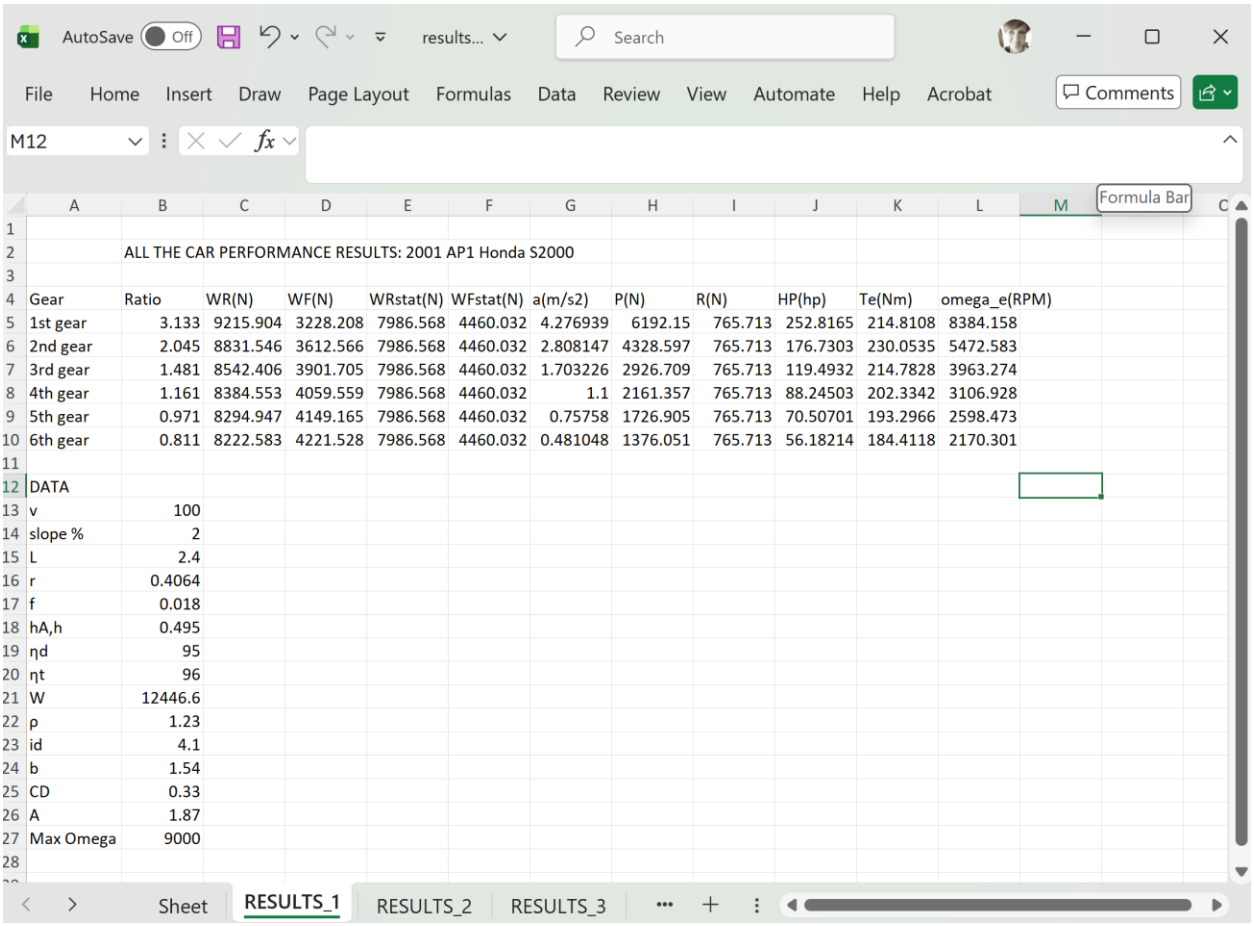


Figure 7 – Possible plot outputs (you can choose to plot anything interesting for you!)



ALL THE CAR PERFORMANCE RESULTS: 2001 AP1 Honda S2000

Gear	Ratio	WR(N)	WF(N)	WRstat(N)	WFstat(N)	a(m/s2)	P(N)	R(N)	HP(hp)	Te(Nm)	omega_e(RPM)
1st gear	3.133	9215.904	3228.208	7986.568	4460.032	4.276939	6192.15	765.713	252.8165	214.8108	8384.158
2nd gear	2.045	8831.546	3612.566	7986.568	4460.032	2.808147	4328.597	765.713	176.7303	230.0535	5472.583
3rd gear	1.481	8542.406	3901.705	7986.568	4460.032	1.703226	2926.709	765.713	119.4932	214.7828	3963.274
4th gear	1.161	8384.553	4059.559	7986.568	4460.032	1.1	2161.357	765.713	88.24503	202.3342	3106.928
5th gear	0.971	8294.947	4149.165	7986.568	4460.032	0.75758	1726.905	765.713	70.50701	193.2966	2598.473
6th gear	0.811	8222.583	4221.528	7986.568	4460.032	0.481048	1376.051	765.713	56.18214	184.4118	2170.301

DATA	
v	100
slope %	2
L	2.4
r	0.4064
f	0.018
hA,h	0.495
ηd	95
ηt	96
W	12446.6
ρ	1.23
id	4.1
b	1.54
CD	0.33
A	1.87
Max Omega	9000

Figure 8 – Result file with tabs